# Algorithms and software engineering: review



## Programming · Level 4 · Unit check

# In this unit

- How work grows: big O
- Recursion
- Data structures
- Testing like an engineer

# The big ideas

- Big O names how work grows; a better method beats a faster phone
- Recursion needs a base case; memoise repeated work
- Choose structures by their operations: sets and dictionaries in one step
- Test the edges, work test-first, review the code

# Activity: Code review circle

1. Warm-up (3 min): One idea from each lesson.
2. Write (7 min): A small function with five asserts.
3. Review (7 min): Pass it on; add a test and a comment.
4. Fix (3 min): Make every test pass.

# Let’s discuss

1. Which idea from this unit will you use in your next project?
2. Which question surprised you, and why?
3. Where would big O matter in a real app you use?

# Exit ticket

- One habit I will keep as a programmer